is called sex linked inheritance. Haemophilia, colourblindness, muscular dystrophy are some examples for X linked inheritance in human beings.

Pedigree analysis is the study of traits as they have appeared in a given family line for several generations. The genetic disorders are of two types- Mendelian and chromosomal. Alternations or mutation in single gene causes Mendelian disorders like, thalassemia, albinism, phenylketonuria, and Huntington's chorea. Chromosomal abnormalities arise due to chromosomal non-disjunction, translocation, deletion, duplication and inversion. Downs syndrome, Klinefelter's syndrome, Turner's syndrome and Patau's syndrome are some of the chromosomal disorders. Downs syndrome is due to trisomy of chromosome 21. Presence of trisomic condition of chromosome 13 results in Patau's syndrome. In Turner's syndrome the sex chromosome is XO and in Klinefelter's syndrome the condition is XXY. An idiogram refers to a diagrammatic representation of chromosomes.

The cytoplasmic extra nuclear genes have a characteristic pattern of inheritance which does not resemble genes of nuclear chromosomes and are known as Extrachromosomal/ Cytoplasmic inheritance. The betterment of human race can be achieved by methods like Eugenics, Euthenics and Euphenics.

Evaluation

T71J6

1. Haemophilia is more common in males because it is a
   a) Recessive character carried by Y-chromosome
   b) Dominant character carried by Y-chromosome
   c) Dominant trait carried by X-chromosome
   d) **Recessive trait carried by X-chromosome**

2. ABO blood group in man is controlled by
   a) Multiple alleles
   b) Lethal genes
   c) Sex linked genes
   d) Y-linked genes

3. Three children of a family have blood groups A, AB and B. What could be the genotypes of their parents?
   a) $I^A I^B$ and $I^o I^o$ b) **$I^A I^o$ and $I^B I^o$**
   c) $I^B I^B$ and $I^A I^A$ d) $I^A I^A$ and $I^o I^o$

4. Which of the following is not correct?
   a) Three or more alleles of a trait in the population are called multiple alleles.
   b) A normal gene undergoes mutations to form many alleles.
   c) **Multiple alleles map at different loci of a chromosome.**
   d) A diploid organism has only two alleles out of many in the population.

5. Which of the following phenotypes in the progeny are possible from the parental combination AxB?
   a) A and B only
   b) A,B and AB only
   c) AB only
   d) **A,B,AB and O**

6. Which of the following phenotypes is not possible in the progeny of the parental genotypic combination $I^A I^O$ x $I^A I^B$?
   a) AB **b) O**
   c) A d) B

7. Which of the following is true about Rh factor in the offspring of a parental combination DdxDd (both Rh positive)?
   a) All will be Rh-positive
   b) Half will be Rh positive
   c) About ¾ will be Rh negative
   d) About one fourth will be Rh negative

8. What can be the blood group of offspring when both parents have AB blood group?
   a) AB only **b) A, B and AB**
   c) A, B, AB and O d) A and B only

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